

Earth Observation Satellite Mission and Systems Design

Preliminary Design Review

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Review stage	Phase 10 concept baseline
Date	25 September 2026
Software	Python 3, NumPy, Matplotlib and automated tests

Purpose. This report records the selected preliminary design, the evidence generated across ten project phases, the final requirements status and the verification work that remains before any flight-design claim could be made.

Portfolio engineering study. Not flight-qualified hardware or guaranteed mission performance.

Executive summary

The project defines a preliminary two-spacecraft multispectral Earth-observation mission for representative UK flood-monitoring use. Ten linked phases connect image requirements to payload sizing, radiometry, orbit and revisit, mission operations, spacecraft subsystem budgets, configuration, uncertainty propagation and final design review.

The final concept uses two spacecraft in a 550 km sun-synchronous orbit, a 10 m ground-sampling distance and an 81.92 km swath. A four-site polar ground-network concept reduces the modeled worst-case priority-delivery latency from 7.00 h to 1.39 h. Strengthened subsystem allocations raise simultaneous budget closure from 10.02% for the initial baseline to 98.48% in the reproducible 5,000-case Monte Carlo study.

The preliminary design review records 15 passed concept-level requirements, no failed requirements and one open item. Three-year mission life remains open because radiation, reliability, parts derating and lifetime testing are outside the current analytical model. The design is therefore accepted as a portfolio-level preliminary concept, not as a flight-qualified system.

Review measure	Result	Status
Cumulative automated tests	61 passed	Verified
Final simultaneous budget closure	98.48%	Pass against 95% target
Worst priority-delivery latency	1.39 h	Pass against 3 h target
Requirements verification	15 pass, 1 open, 0 fail	Conditional acceptance

Mission definition

The mission is an engineering reference case intended to demonstrate end-to-end space-systems reasoning. It is not tied to a customer contract. The top-level performance targets are spatial resolution, swath, revisit, delivery latency and a 60 kg wet-mass limit per spacecraft.

Parameter	Selected concept	Design rationale
Architecture	Two spacecraft	Improves representative UK revisit
Orbit	550 km sun-synchronous	Balances coverage, resolution and lighting
Inclination	97.593 deg	Approximate J2 sun-synchronous solution

Imaging	10 m GSD, 81.92 km swath	Meets mission targets
Spectral bands	Blue, green, red and NIR	Flood and vegetation context
Ground segment	Four conceptual polar sites	Closes preliminary 3 h delivery target

Engineering method

The analysis uses transparent first-order physical models linked through common design variables. Each phase produces machine-readable outputs and automated tests. The workflow is deliberately traceable: requirements drive sizing; sizing drives subsystem budgets; uncertainty analysis identifies weak margins; the final review strengthens allocations and rechecks closure.

Phase	Analysis	Principal evidence
1 to 3	Payload geometry, radiometry and optimisation	Focal length, aperture, SNR, mass trade
4	Orbit and coverage	SSO inclination, ground track, representative revisit
5	Operations and ground access	Capacity, storage and latency
6 to 7	Subsystem sizing and integration	Mass, power, link, ADCS, propulsion and thermal budgets
8	Spacecraft configuration	Bus envelope, component layout, centre of mass and STL concept
9	Monte Carlo robustness	5,000 cases and global sensitivity ranking
10	Final preliminary design review	Selected allocations and final verification matrix

Payload design

An 8,192-pixel detector with 10 micrometre pitch gives a 550 mm focal length at 550 km altitude for 10 m nadir sampling. The resulting cross-track swath is 81.92 km. Radiometry and image-smear constraints were traded against aperture and exposure time. Phase 3 selected a 160 mm

preliminary aperture; the final robustness allocation uses a 190 mm aperture proxy to protect worst-band SNR.

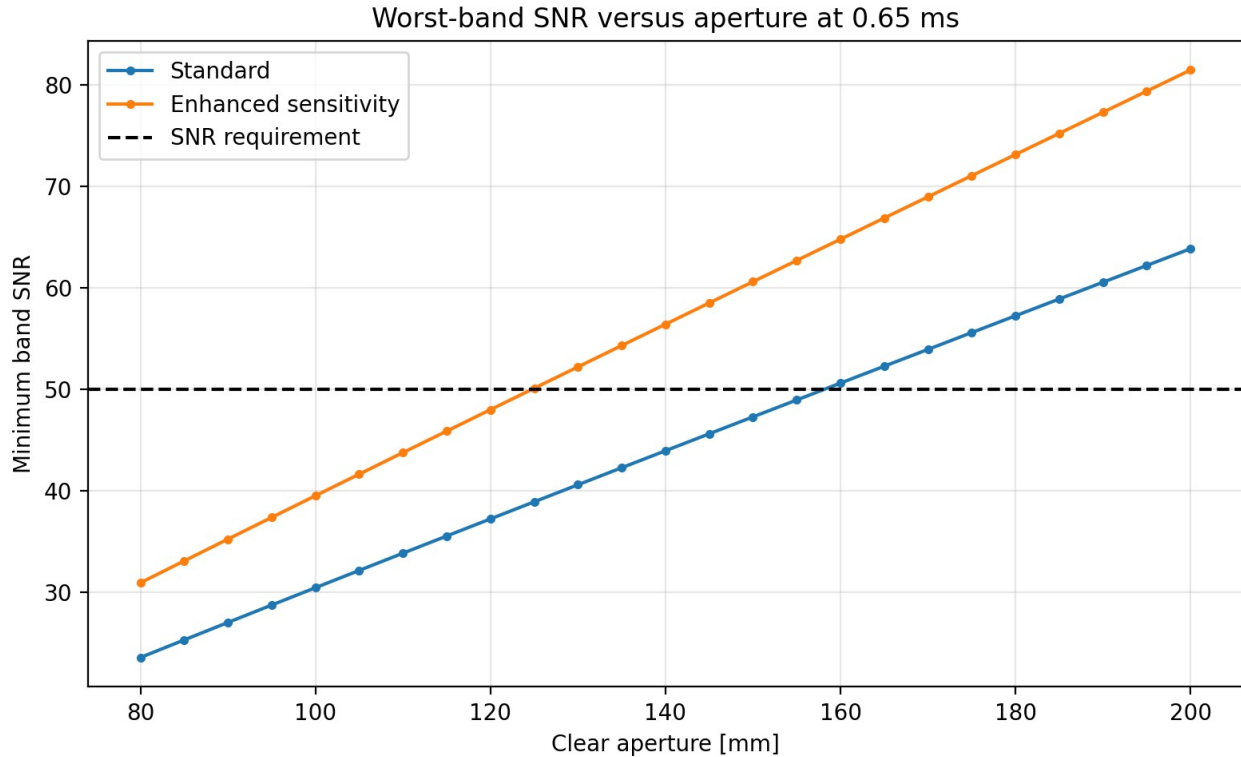


Figure 1 Aperture and worst-band SNR optimisation

The aperture scaling remains a response-surface approximation. A detailed optical design must confirm modulation transfer function, stray light, focus stability, detector selection, baffle geometry and the mass and alignment effects of the larger telescope.

Orbit and coverage

The orbit model propagates circular Earth-fixed ground tracks and calculates the retrograde inclination required for approximate sun-synchronous nodal precession. A two-spacecraft architecture at 550 km with up to 25 deg off-nadir access produced a 37.79 h worst revisit among five representative UK sites, meeting the 48 h target.

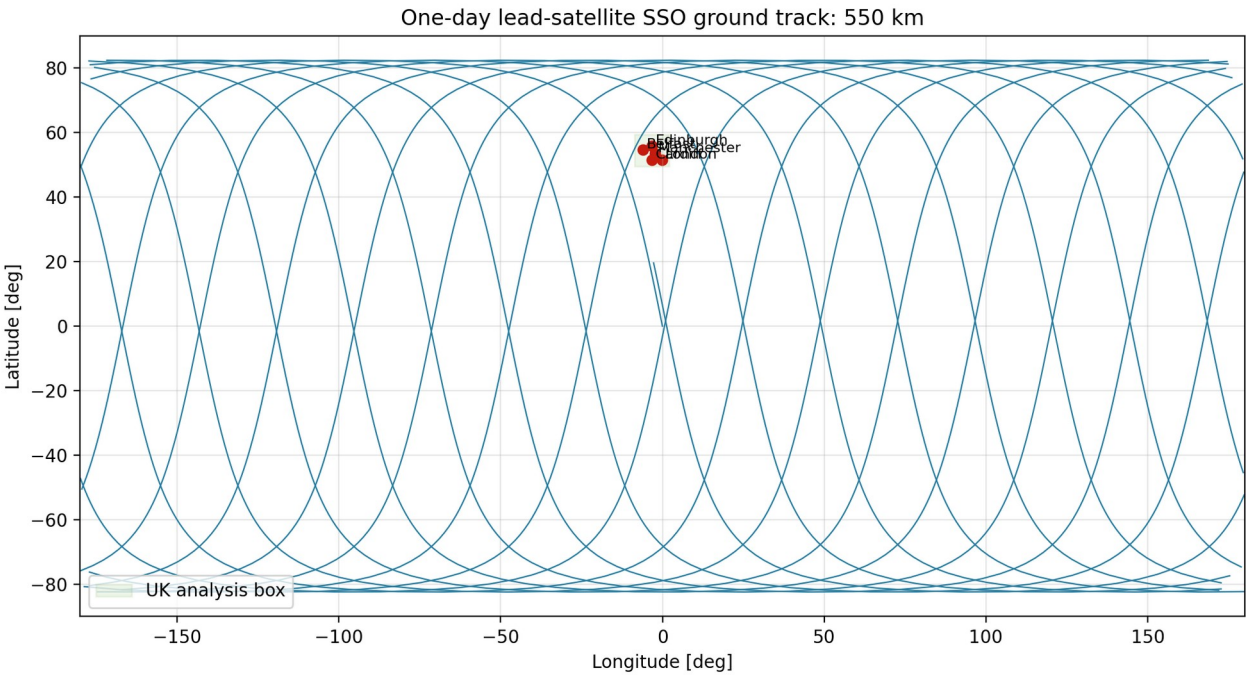


Figure 2 Preferred two-spacecraft ground-track representation

The revisit result applies to five sites rather than a complete UK-area coverage mask. Future mission analysis should use higher-fidelity propagation, cloud statistics, imaging constraints, exclusion zones and target-priority scheduling.

Mission operations and communications

Four representative 1,000 km scenes generate 39.32 Gbit per spacecraft per day after the assumed 4:1 compression. The original Goonhilly and Svalbard network had ample average capacity but a 7.00 h modeled worst delivery latency. Phase 10 adds conceptual Alaska and Antarctic polar sites; the idealised geometry gives 1.39 h worst latency and 1,163.09 Gbit/day mean capacity per spacecraft.

Ground architecture	Capacity per spacecraft	Worst latency	3 h target
Goonhilly	143.37 Gbit/day	9.94 h	Fail
Goonhilly and Svalbard	562.28 Gbit/day	7.00 h	Fail
Four-site polar concept	1,163.09 Gbit/day	1.39 h	Pass

This is an access-geometry result, not a contracted service assessment. It excludes simultaneous antenna conflicts, weather, maintenance, regulatory licensing, network routing, product processing time and commercial availability.

Spacecraft subsystem design

Phase 6 sized the initial power, battery, communications, attitude-control, propulsion, thermal and mass budgets. Phase 9 showed that several deterministic margins were too narrow under uncertainty. The final allocation therefore increases capability in the dominant risk areas.

Subsystem	Calculated baseline need	Final allocation
Wet mass	55.29 kg with system margin	60 kg limit
End-of-life solar power	130.35 W	190 W
Battery	53.47 Wh	80 Wh
Storage	2.95 GB preliminary	16 GB
X-band link	3.86 dB residual margin	+4 dB design improvement
Reaction-wheel torque	0.00970 N m	0.018 N m
Reaction-wheel momentum	0.0861 N m s	0.180 N m s
Mission delta-v	48 m/s	70 m/s
Radiator area	0.534 m ²	0.750 m ²

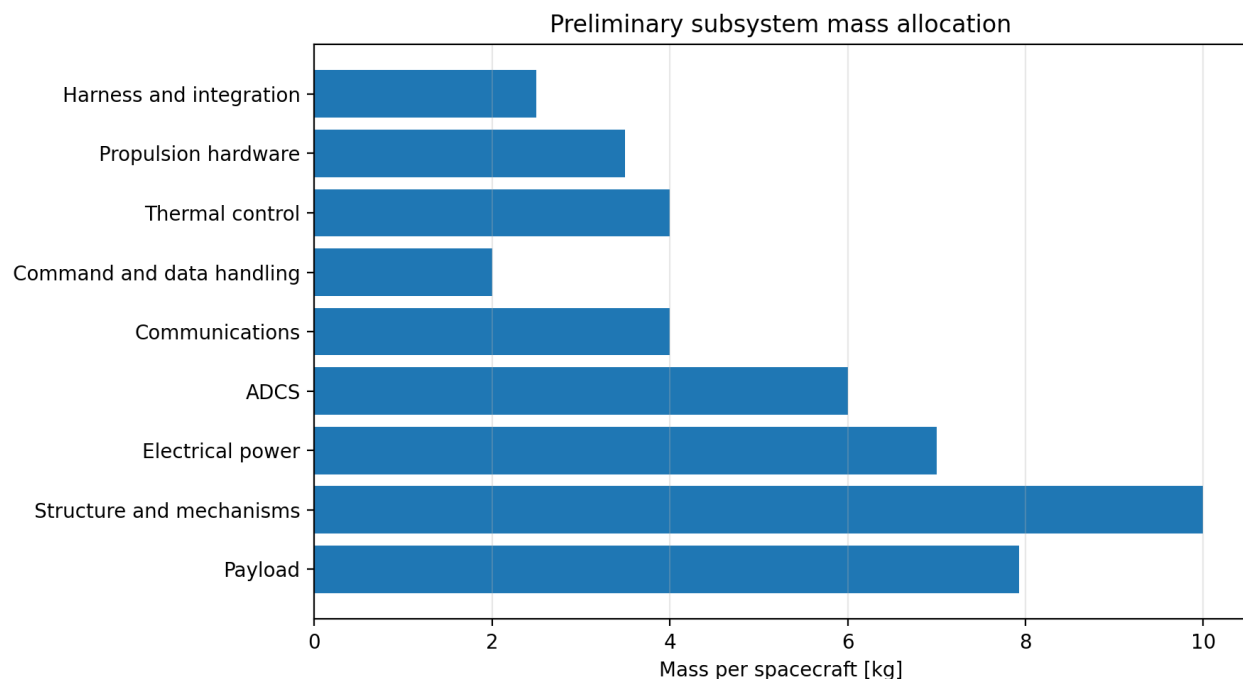
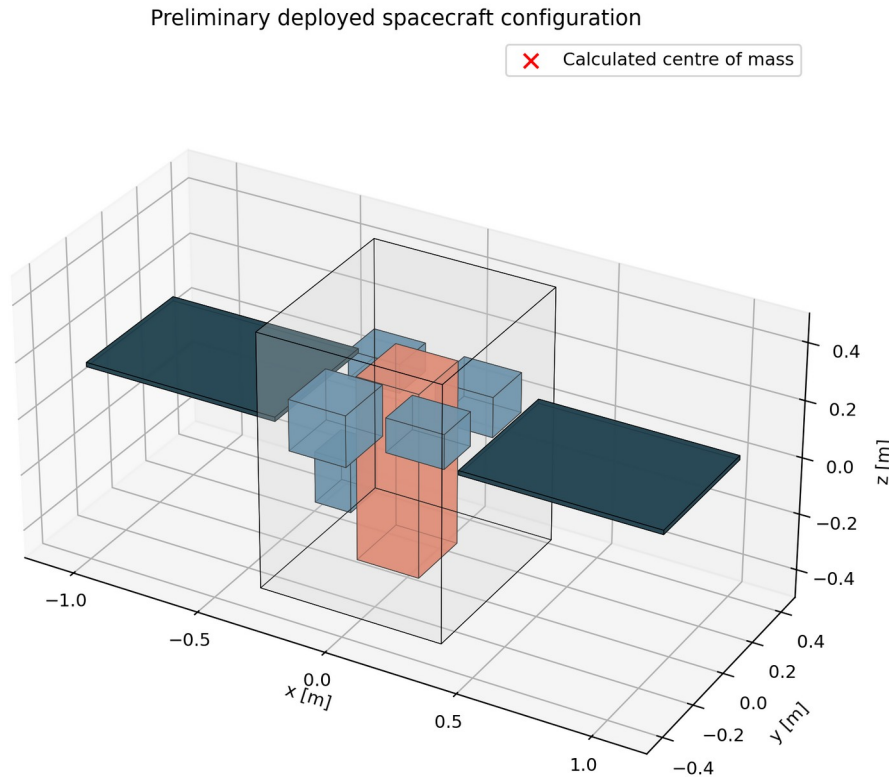


Figure 3 Preliminary subsystem mass budget

Spacecraft configuration

The configuration model places the payload and major subsystems inside a 0.70 m by 0.70 m by 0.90 m bus. The deployed envelope is 2.20 m by 0.70 m by 0.90 m. The preliminary centre of mass lies approximately 19 mm from the bus centre. The configuration allocates 0.75 m² of solar array and 0.60 m² of radiator area in Phase 8; the final 0.75 m² radiator allocation therefore requires a configuration update before detailed design.

**Figure 4 Preliminary spacecraft configuration**

The generated STL is a packaging concept rather than a manufacturing CAD assembly. Structural load paths, mechanisms, harnessing, fasteners, keep-out zones, thermal interfaces and launch-vehicle accommodation remain outside its scope.

Robustness and sensitivity

The deterministic Phase 7 design passed every modeled subsystem budget but had little margin in several areas. Phase 9 propagated explicit uncertainty distributions across payload signal, mass, power, communications, attitude control, propulsion, thermal performance, data volume and ground availability. Only 10.02% of baseline cases passed every budget simultaneously.

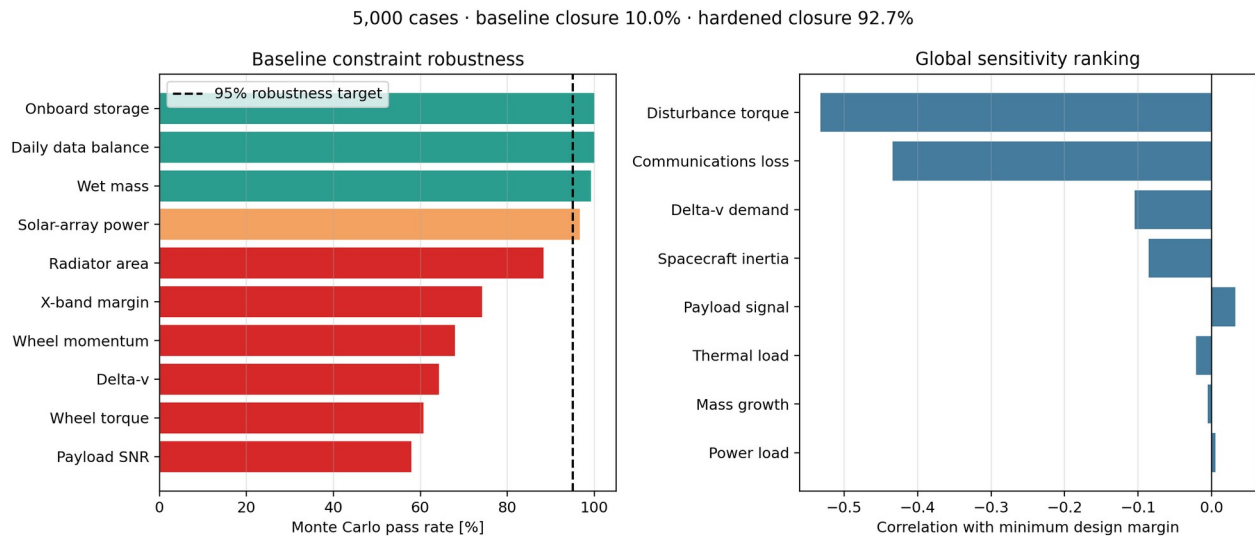


Figure 5 Baseline robustness and sensitivity evidence

The Phase 9 hardened comparison reached 92.66%. The final Phase 10 allocations increase aperture, power, communications, wheel, delta-v and radiator capability further, producing 98.48% simultaneous closure in the same 5,000 deterministic cases. Every individual modeled budget exceeds the 95% pass-rate target.

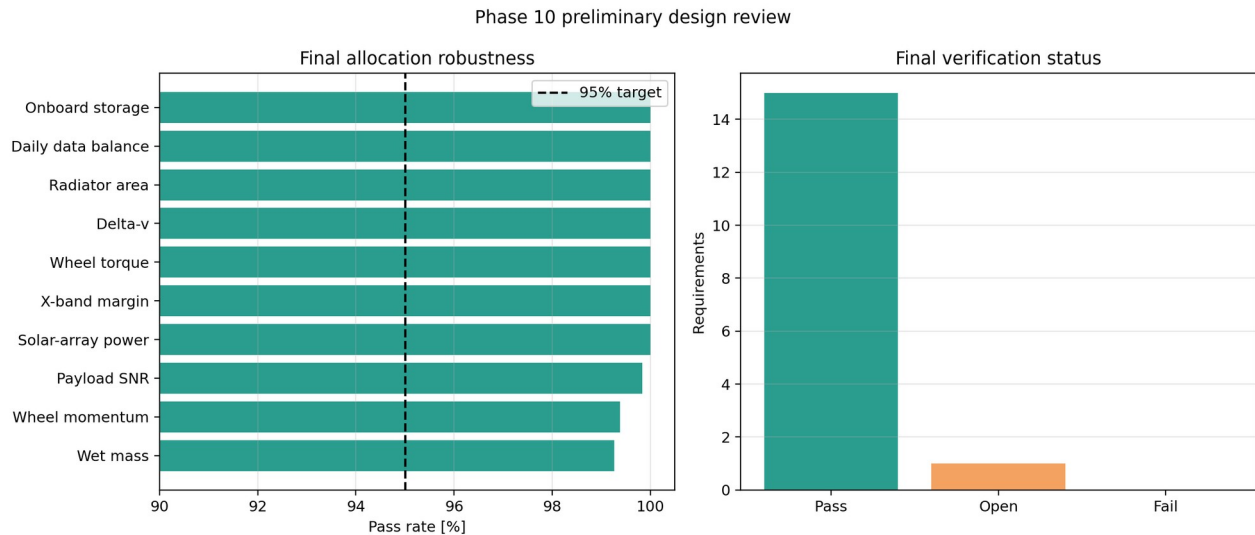


Figure 6 Final allocation robustness and requirements status

Requirements verification

The final matrix contains 16 tracked requirements. Fifteen pass through analysis, simulation or budget evidence. No requirement remains failed. Mission lifetime remains open because the current work does not contain the evidence needed to make a three-year reliability claim.

ID	Requirement	Result	Status
MIS-001	10 m nadir sampling	10.00 m	Pass

MIS-002	At least 50 km swath	81.92 km	Pass
PAY-001	Worst-band SNR at least 50	Final allocation passes	Pass
MIS-003	UK-site revisit within 48 h	37.79 h	Pass
MIS-004	Priority delivery within 3 h	1.39 h	Pass
SYS-001	Wet mass no more than 60 kg	55.29 kg baseline	Pass
ROB-001	At least 95% robust closure	98.48%	Pass
MIS-005	Three-year mission life	Analysis pending	Open

The complete matrix, including data, power, communications, ADCS, propulsion and thermal requirements, is supplied as `final_requirements_matrix.csv` in the Phase 10 results folder.

Verification evidence

The repository contains 61 cumulative automated tests. They check mathematical behavior, physical bounds, expected monotonic trends, reproducibility, output generation, budget closure and deliberate retention of open verification work. Passing tests demonstrate internal implementation consistency; they do not validate vendor hardware or replace environmental testing.

- Reproducible seed 260925 for the 5,000-case robustness study.
- Machine-readable CSV and JSON results for traceability.
- Plots generated directly from the analysis outputs.
- Explicit Pass, Fail and Open states rather than silently suppressing gaps.
- Cumulative unit-test discovery across all ten phases.

Limitations and remaining work

Area	Current limitation	Required next evidence
Optics	Response-surface aperture scaling	Detailed optical prescription, MTF and tolerance model

Coverage	Five representative UK sites	Area coverage, weather and scheduling simulation
Ground segment	Idealised station geometry	Provider availability, conflicts, licensing and cost
Structure	Packaging STL only	CAD assembly, launch-load FEA and modal verification
Thermal	Steady first-order sizing	Hot and cold transient orbital model
Radiation and reliability	Not modeled	Parts analysis, FMEA, derating and lifetime demonstration
Operations	No product-processing chain	End-to-end scheduling and processing latency
Cost and programme	Outside present scope	Cost estimate, schedule and procurement plan

Conclusion

The project demonstrates a complete preliminary space-systems design loop: requirements were translated into payload and orbit choices, spacecraft and ground budgets were integrated, uncertainty exposed weak margins, and the final allocation was revised against measurable closure criteria. The selected concept meets all modeled performance and budget requirements, with 98.48% simultaneous Monte Carlo closure and 1.39 h modeled priority latency.

The correct review decision is conditional acceptance at preliminary concept level. The analytical design is coherent and reproducible, but three-year mission life and the detailed hardware, structural, optical, thermal and operational evidence remain future verification work. This distinction is central to the engineering value of the study.

Repository guide

Command	Purpose
<code>python -m unittest discover -v</code>	Run all 61 cumulative tests
<code>python final_design_review.py</code>	Regenerate final review data and figure
<code>python build_final_report.py</code>	Regenerate this DOCX report